Hardware Assembly & Integration — Progress Report

Three-phase pump power abnormality monitoring and protection · Personal engineering portfolio

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1. Project overview

This report records an earlier integration milestone for a low-voltage embedded prototype that measures three phase-current signals and three phase-voltage signals. The design uses the ESP32 for control and voltage-channel acquisition, and the ADS1115 for current-channel acquisition. It describes the intended protection concept; it does not claim that the final system has been validated on a live pump.

2. Hardware platform

Category	Selected component / configuration	Role / notes
Controller	ESP32-WROOM-32	Control logic, I²C peripherals and voltage-channel ADC inputs
External ADC	ADS1115, 16-bit, four-channel	Used for three current-sensor channels
Current sensing	3 × SCT-013, 100 A / 50 mA	Current transformer outputs; 33 Ω burden and bias circuit as recorded
Voltage sensing	3 × ZMPT101B modules	Connected to ESP32 GPIO 35, 32 and 33
Relay output	5 V TONGLING JQC-3FF relay module	Intended to command the contactor-coil interface; requires a reviewed safety design
Display and controls	0.96-inch SSD1306 OLED, 3 LEDs, button	Status display and manual reset
Low-voltage power	ATX 5 V rail, 1 A fuse	Fused and decoupled low-voltage supply
Panel components referenced	Hyundai HGC 25 contactor; HMMS-32K	Recorded as project context; compatibility and interface must be verified

3. Sensor routing decision

The ADS1115 has four single-ended inputs, while the prototype needs six measurement channels. The recorded decision is to use the ADS1115 for the three current channels and the ESP32 internal ADC for the three voltage channels, avoiding a second external ADC. The trade-off is lower resolution and potentially greater noise on the voltage readings; filtering and calibration were identified as mitigation steps.

Signal	Sensor	Connection	Acquisition
Current — Phase 1	SCT-013, burden/bias	ADS1115 AIN0	ADS1115
Current — Phase 2	SCT-013, burden/bias	ADS1115 AIN1	ADS1115
Current — Phase 3	SCT-013, burden/bias	ADS1115 AIN2	ADS1115

Signal	Sensor	Connection	Acquisition
Voltage — Phase 1	ZMPT101B	ESP32 GPIO 35	ESP32 internal ADC
Voltage — Phase 2	ZMPT101B	ESP32 GPIO 32	ESP32 internal ADC
Voltage — Phase 3	ZMPT101B	ESP32 GPIO 33	ESP32 internal ADC

4. ESP32 pin map

ESP32 pin	Function
VIN / 5V	Fused ATX +5 V input
GND	Common low-voltage ground, subject to circuit review
GPIO 21 (SDA)	I²C data for OLED and ADS1115
GPIO 22 (SCL)	I²C clock for OLED and ADS1115
GPIO 25	Relay control output
GPIO 26	Manual reset button
GPIO 27	Red fault LED
GPIO 14	Yellow warning LED
GPIO 12	Green normal LED
GPIO 35	Voltage sensor 1 (ZMPT101B)
GPIO 32	Voltage sensor 2 (ZMPT101B)
GPIO 33	Voltage sensor 3 (ZMPT101B)
3.3 V	OLED and current-sensor bias circuits, as recorded

5. Assembly status recorded on 28 September 2026

Item	Reported status
Low-voltage power supply and 1 A fuse	Completed and verified
ESP32, OLED, LEDs, relay and button	Completed and tested
Three current sensors and burden/bias circuits	Completed and verified
Three voltage sensor modules	Completed
ADS1115 and six-channel routing	Completed and verified

6. Remaining work at the time of this report

- Complete RMS calculation, protection logic, state-machine behavior and trip handling.
- Calibrate all current and voltage channels.
- Review and obtain approval for any relay connection to the contactor-coil circuit.
- Complete staged testing and document evidence before any live-pump test.
- Prepare final code, documentation and portfolio repository.

7. Scope and limitations

This is a historical integration snapshot dated 28 September 2026. It is superseded for later implementation status by the separate report dated 3 October 2026. It describes the embedded low-voltage prototype, not the complete power/control panel. Any work on a live panel or pump requires separate equipment-owner approval and qualified personnel. No company endorsement, installation permission, certification or live-test result is asserted.

Revision note: standardized sensor naming and role wording; removed job-title inconsistency and company-branded footer; distinguished reported assembly status from validated performance.